Language Identification for Non-Standard Varieties

Language Varieties as Geolects on a Geodesic Grid

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Background

Background

- Language identification (LID) → filtering corpora for pretraining LLMs (Oepen et al., 2025)
- Solved for standard varieties, **not for non-standard ones** (Caswell et al., 2020)
- It assumes a **fixed inventory** of discrete labels with names and clear boundaries
- Non-standard varieties have **no agreed name** and often no ISO code!
- The **label set itself** is the bottleneck

Let's Redefine LID!

"Drmit ischsi o für vili Bürgerlechi wäubar."

(This also makes her electable for many bourgeois voters.)

What language variety is this?

↓

Swiss German (`gsw`)

reframe

Where on earth could this text have been produced?

⇒ a **geolect**

↓

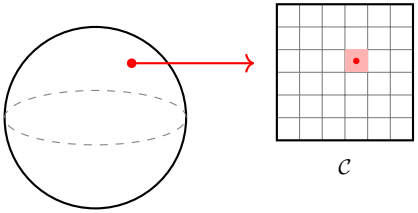
● **Location** (***Bern***)

The Idea: A Geodesic Grid

- Discretize the earth into a grid of **square cells** $\mathcal{C}$
- Given grid resolution g (degrees):

$$x = \left\lfloor \frac{\lambda + 180}{g} \right\rfloor, \quad y = \left\lfloor \frac{\phi + 90}{g} \right\rfloor$$

- A text $\rightarrow$ a **cell**, not a float
- Classification over cells, not labels
- Every prediction is a **location**: cells map directly to coordinates



Related Work in Brief

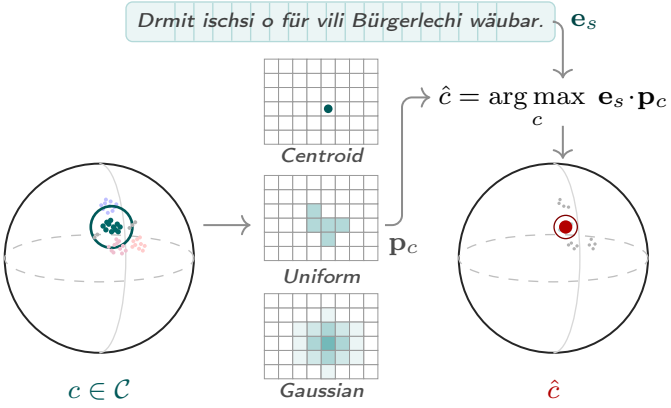
- Document classification into grid cells with bag-of-words features (Wing and Baldridge, 2011; Roller et al., 2012)
- Smoothing sparse grid counts with Gaussian kernels for fine-grained cells (Hulden et al., 2015)
- Neural classifiers for tweet geolocation (Miura et al., 2017)
- Geolocating speech from radio broadcasts via self-supervised representations (Foley et al., 2024)
- Interpolating ASR performance across Italian dialect areas using geographic proximity (Shim and Plank, 2025)

The gaps we address:

Prior work is mostly **single-language** and often **speech**. We bring the continuous, geographic perspective to **text** across **multiple language families**, with geography **at the center** of the model.

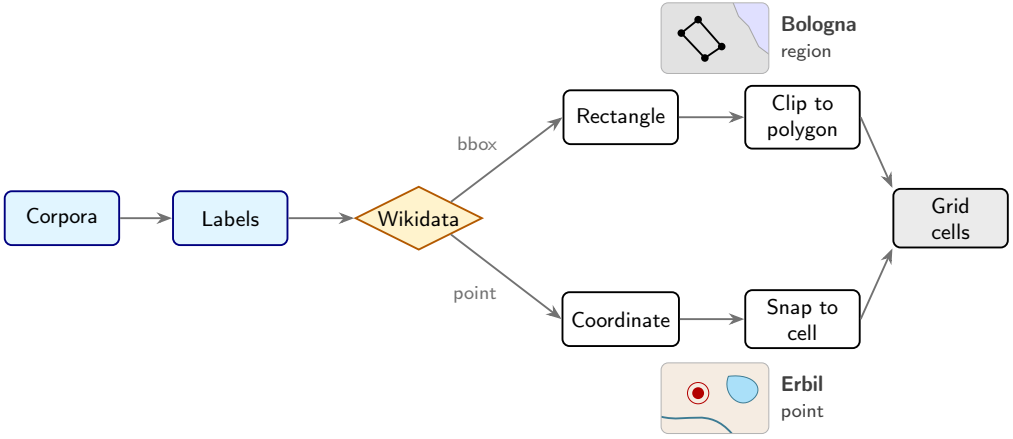
Methodology

Approach

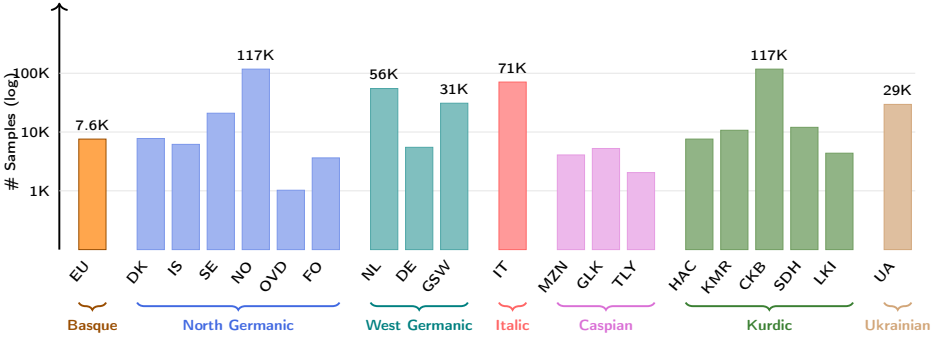

Drmit ischsi o für vili Bürgerlechi wäubar.
$\mathbf{e}_s$
Centroid
Uniform
Gaussian
$\mathbf{p}_c$
$\hat{c} = \arg\max_c \ \mathbf{e}_s \cdot \mathbf{p}_c$
$c \in \mathcal{C}$
$\hat{c}$

Data: Geographic Anchoring

Location labels are messy, and oftentimes lack geographic coordinates



Data: GeoVar

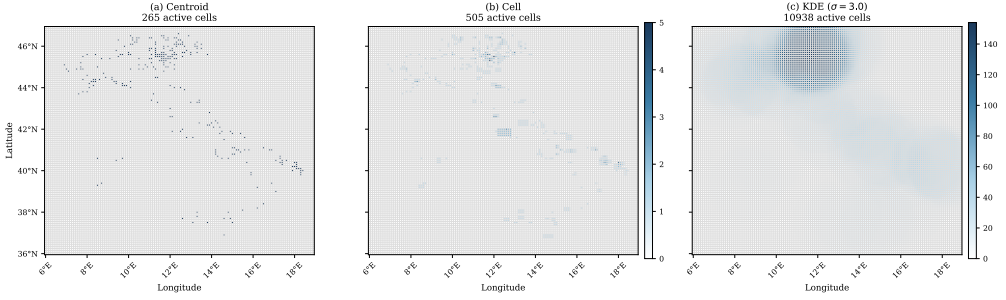


- **20** languages
- **929** geolects
- **517,985** samples
- **7** continua

Geolect–Cell Mapping

Three strategies forming a spectrum of increasing spatial smoothness:

- **Centroid**: each geolect maps to a single cell at the center of its region
- **Uniform**: embeddings are distributed uniformly across all cells in the geolect's region (hard boundaries)
- **Gaussian**: an isotropic kernel spreads mass outward from the centroid with gradual decay beyond the region boundary



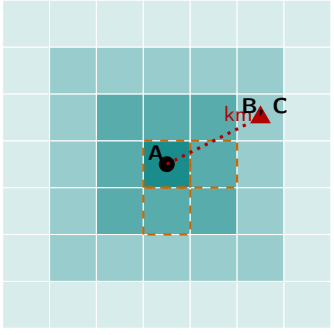
Classification Methods

- All methods use **CANINE-C**, a character-level encoder
- **Baselines**: no learning on the grid
  - **B1**: logistic regression on variety labels, no grid
  - **B2**: nearest centroid, i.e. the closest cell prototype
- **Learnable methods**, from simple to expressive
  - **M1**: logistic regression over cells instead of labels
  - **M2**: ProtoNet, learns a projection that pulls same-cell sentences together
  - **M3**: fine-tunes the encoder, then predicts by nearest prototype
- One model per dialect continuum

Methodology: Evaluation

- **Variety identification**: accuracy ↑, macro-F1 ↑, cell accuracy ↑
- **Spatial localization**: median error in km ↓, Acc@Nc ↑

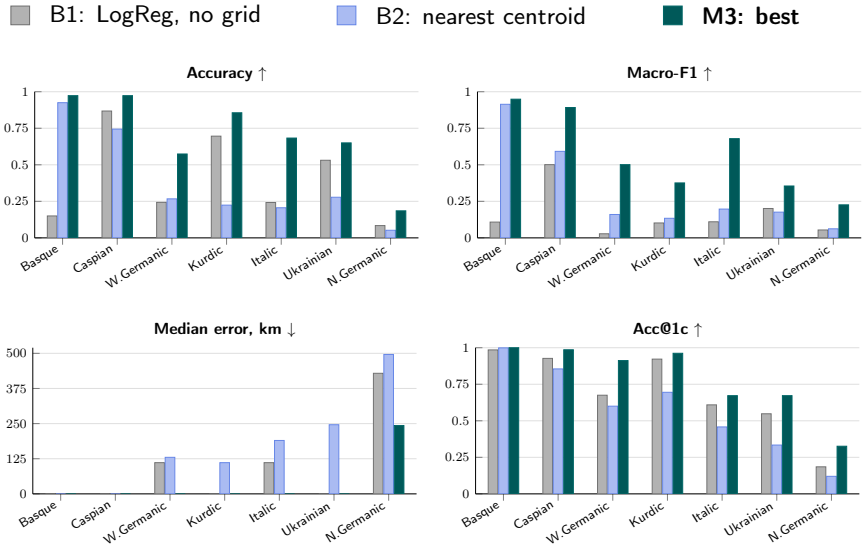
Coarser grids help the first and hurt the second ⇒ we report both



- Acc@0c: exact cell
- Acc@1c: within 1 cell
- Acc@2c: within 2 cells
- Acc@3c: within 3 cells
- ● true cell (centroid)
- ▲ predicted cell; dotted = km error
- region R_d: cell accuracy
- [A] varieties in a cell: correct if the true one is in $\hat{c}$

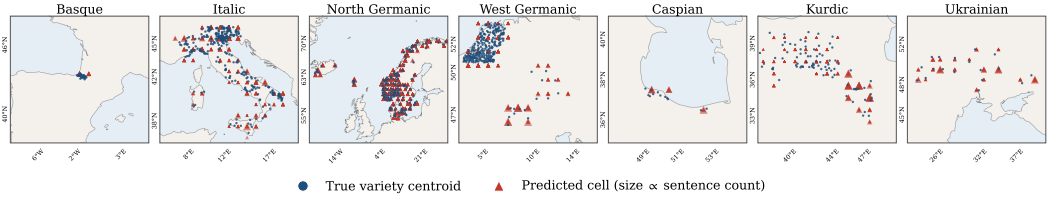
Results

Can a Single Sentence Tell Us Where It Comes From?



Ordered by geographic dispersion σ_{geo}

Why is North Germanic so much harder than Basque?



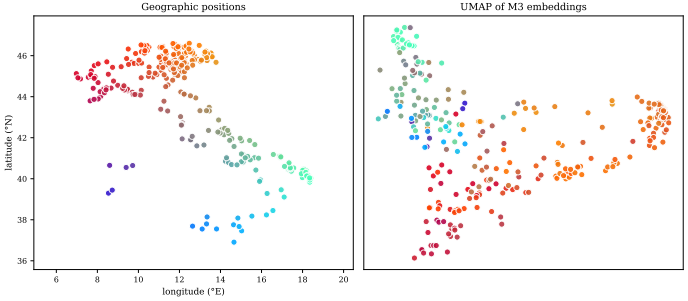
- Difficulty tracks **how far a continuum spreads**, not how many varieties it holds
- More varieties sharing a **cell** go with higher F1, as there are fewer cells to tell apart
- Per continuum:
  - **Basque**, **Caspian**: compact, above .97 accuracy
  - **North Germanic**: most dispersed, 243 km off
  - **Kurdic**: high accuracy, low F1, as Central Kurdish dominates the data

Which Modelling Choices Actually Matter?

- **Grid size**: coarser cells raise accuracy; every continuum lands on 1°
- **Mapping**:
  - Centroid: best on all seven continua
  - Uniform: same as centroid for most continua, since at 1° most regions fit in a single cell
  - Gaussian: inflates accuracy and F1 by pooling more varieties per cell, but exact-cell accuracy collapses
- **Merging continua?**: barely needed!
  - One model on Italic, West Germanic and North Germanic (766 varieties): accuracy barely drops
  - Only 59 of 19,569 errors cross a continuum, just 26 across the Alps
- **Encoder**: character-level CANINE-C beats subword encoders (LaBSE, MiniLM)

Does the Model Rediscover the Varieties Map?

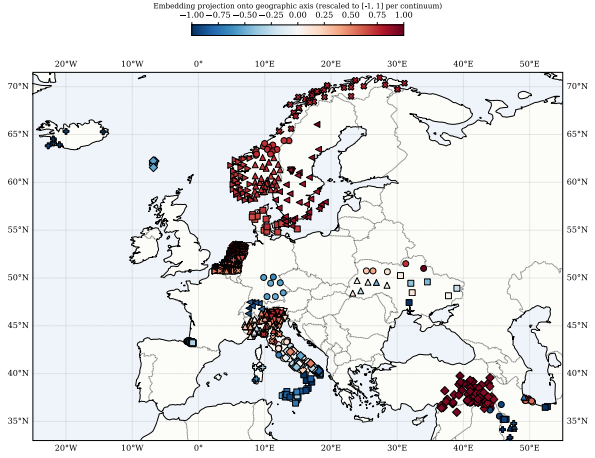
- If variation is gradual, neighbouring varieties should also be neighbours in the embedding
- They are: the geographic gradient is preserved
- Sardinia and the southern islands sit apart as separate clusters
- Both regimes coexist ⇒ a continuum with clusters



Italic: geographic position (left) vs. embedding (right); same colour = same geolect

Does Every Continuum Leave a Geographic Trace?

- If geography is encoded, a variety's location should be readable from its embedding
- It is the case in Italic, both Germanic groups and Kurdic
- Less so in Basque (too compact), Caspian and Ukrainian (too few varieties)
- Geography is encoded wherever the continuum spans real distance



Conclusion

Conclusion

Identify varieties by **where they are attested**, not by a fixed label list

- Circumvents the **labelling bottleneck** we started with: a variety needs a location, not a name
- GeoVar: **929 geolects** in 20 languages and 7 continua, anchored to coordinates

  Openly available at `github.com/sinaahmadi/GeoVar`



- Difficulty depends on **how far varieties spread**, not how many there are
- The model learns the map: geography and continuum boundaries emerge from the data
- **What's next**:
  - Geography as a training signal
  - Learned location encoders and equal-area grids (e.g. H3)
  - Coverage beyond Europe and the Middle East

Thank you! Questions?

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